Bounce-apart calculations

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use momentum and energy to check an elastic collision.

SUCCESS CRITERIA - I can...

1. Write a signed momentum balance.
2. Check kinetic energy separately.
3. Explain why a collision cannot be elastic if its total kinetic energy changes.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “elastic” and “conserved” to explain your answer. Sentence starters are optional.

elastic: a collision that conserves total kinetic energy

conserved: unchanged in total under the stated conditions

kinetic energy: energy associated with motion

01 Warm-up: bring back an earlier idea

Two **3 kg** carts latch together after colliding. Is this perfectly inelastic? **State what remains conserved** if external impulse is negligible.

Word help: inelastic = a collision that transforms some kinetic energy to other forms

Sentence starter: I notice … . This supports … because … .

Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: $p_{total,\,i} = p_{total,\,f}$; $K_{total,\,i} = K_{total,\,f}$.
Isolated elastic collision; $K = \frac{1}{2}mv^2$. Equal masses exchange velocities in the ideal one-dimensional case.

Worked reasoning: A **1 kg** cart moving at **+3 m/s** hits a stationary **2 kg** cart. The first rebounds at **-1 m/s**. Total momentum was **+3 kg·m/s**, so the second must have momentum **+4 kg·m/s** and velocity **+2 m/s**.

Your turn: Draw momentum arrows before and after; verify the total.

Sentence starter: In my work, I used … to show … . I labeled it … .

Before	After

Label mass, velocity and momentum.

Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

Equal-mass carts collide head-on elastically. A starts right at **3 m/s**; B rests. **Sketch the after state**.

Sentence starter: In my work, I used … to show … . I labeled it … .

Before contact: add velocity arrows.



Marzano check - circle: ① ② ③

04 Apply the physics

A **1 kg** cart approaches a bumper at **5 m/s** right and rebounds at **1 m/s** left. Take right as positive. **Label both velocities** and **calculate the cart’s momentum change**.

Sentence starter: I found … [unit]. This value means … in this situation.

Before	After

Label objects and signed velocities.

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Explain your reasoning

A pair’s total momentum matches before and after, but kinetic energy falls. Is the collision elastic? Explain **0**.

Sentence starter: I think … . I can point to …, which shows …, so … .

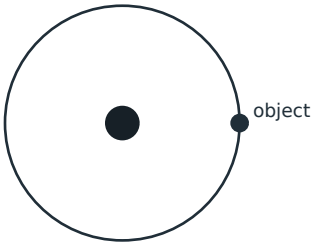
Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A satellite follows a circular orbit at a constant **6 km/s**. **Draw velocity** and acceleration at the top and side of the orbit. **Explain the different directions**. **Label velocity** and inward acceleration on the circular-path diagram.

Word help: orbit = a path around another body under gravity

Sentence starter: Both … . They differ in … because … .



Marzano check - circle: 1 2 3

07 Exit check

Equal-mass carts have initial velocities **+4** and **-1 m/s**. In an ideal **1** D elastic collision, what are final velocities? Organize the known quantities on the before-and-after diagram.

Word help: ideal = using stated simplifying conditions

Sentence starter: I notice … . This supports … because … .

Before	After

Label objects and signed velocities.

Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

A **2 kg** cart starts at **3 m/s** and strikes a stationary **1 kg** cart. If the heavy cart finishes at **1 m/s** right, find the light cart's velocity. Organize the known quantities on the before-and-after diagram.

Sentence starter: I found … [unit]. This value means … in this situation.

Before	After

Label objects and signed velocities.

Marzano check - circle: 1 2 3